Hugo Larios

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Summary:

Mechanical Engineering senior at Cornell University with hands-on experience across propulsion, mechatronic design, and aerodynamic analysis. Progressed from propulsion team member to subteam lead on Cornell's Design Build Fly team, building electrical systems and a static thrust rig for competition aircraft; currently applies FEA and fluid dynamics modeling as a member of the Analytics subteam. Proficient in SolidWorks, Fusion 360, ANSYS (Workbench/Fluent), and Python/C/C++/Java/MATLAB. Available to relocate for summer 2027.

Education:

Cornell University, College of Engineering Ithaca, NY

B.S. Mechanical Engineering, GPA 3.33/4.0 *Expected May 2027*

Continuing to early Aerospace M. Eng. Program at Cornell University *Expected December 2027*

Relevant Coursework:

Propulsion of Aircraft & Rockets, Intermediate Fluids, System Dynamics, Heat Transfer, and Mechanics of Materials.

Project Team Experience:

Cornell Design Build Fly (CUDBF) Ithaca, NY | *September 2024-Present*

Analysis Subteam Member *May 2026-Present*

- Modeled a 14x7e APC propeller on SolidWorks, then simulated airflow past it using ANSYS Fluent (CFD), generating aerodynamic data used across the team to compare and select propeller designs.
- Serves as senior advisor for the Propulsion Subteam, supporting propulsion system design and analysis.

Propulsion Subteam Lead *May 2025-May 2026*

- Led a 7-person subteam, organizing weekly tasks and dependencies via Slack and Notion to meet design milestones, preventing uncommunicated design overlaps between subteams.
- Utilized a Python and MATLAB sizing tool to model and optimize motor-propeller-battery selections, validating over sixty predictions with experimental thrust data through iterative testing and statistical analysis.
- Designed three test procedures and checklists for propulsion system testing to improve repeatability, support safe operations for current and future team.
- Modeled integration components and produced drawings, parts lists, and manufacturing documentation for the end-of-year design report.

Propulsion Subteam Member *September 2024-May 2025*

- Integrated a flight controller and electrical system into two hand-built gliders; used Mission Planner and Ardupilot to automate flight paths and correct control-surface movement.
- Applied machining, crimping, and soldering techniques to assemble electronic components, ensuring high-quality material flow and assembly.
- Assisted with prototype fabrication and assembly of electronic subsystems, ensuring tight tolerances and reliable follow-through on design specifications.

Notable Projects:

Spider Robot, (Inspired by the game TUNIC) *June 2026-Present*

- Designed and modeled a 12-degree-of-freedom quadruped robot in Fusion 360, developing separate mechanical and computational subsystems.
- 3D-printed a PLA set of prototype legs, interfacing servos with a Raspberry Pi and PCA9685 PWM driver to verify joint actuation and walking motion.
- Created a full circuit diagram and selected all electrical components needed to support a 12-servo actuation and control system.

Static Thrust Rig, (Gathers Propulsion Data) *September 2025-March 2026*

- Designed and built a static thrust test rig with a 7-person team, integrating an HX711 load cell and Arduino to measure and verify thrust output across twenty battery-motor-propeller combinations against theoretical models.
- Expanded instrumentation to include voltage, current, RPM, and throttle sensors for full propulsion system characterization.
- Interfaced Arduino with Excel for automated data logging, enabling direct comparison of test results to inform propulsion selection for the competition aircraft.

Skills & Languages:

CAD & Analysis: SolidWorks, Fusion 360, eCalc, ANSYS Workbench, ANSYS Fluent, LabVIEW, XFLR5

Programming: Python, C, C++, Java, MATLAB

Manufacturing: Drill Press, Milling, Lathing, 3D Printing, Laser Cutting, Crimping, Soldering

Languages: Fluent in English, Intermediate French, Intermediate Spanish